

Senus PLC — AI-Native Board Report

Assiduous Technology Graduate Assessment · Sean Gavigan

100% / 100%

extraction precision / recall
vision path: 44/44

1.3 mo

cash runway at 30 Jun 2026
the board-level finding

100

deterministic metrics computed
zero calculated by the AI

3

source-document defects found
by automated reconciliation

What this is

A board reporting platform for Senus PLC (Euronext Access Dublin, SENUS) built from 15 published source documents. An AI pipeline extracts every financial figure with page-level provenance into Postgres; every board-facing number is then computed by deterministic Python, never by the model; and AI-generated commentary and live Q&A are grounded strictly in that computed data and a management transcript, never in the raw filings.

The core engineering problem

The audited FY2025 accounts — the only source for full-year figures — are a scanned, 90°-rotated photograph with zero machine-readable text. Other documents are normal text PDFs. The pipeline measures each document's text density and routes it down a vision or text extraction path accordingly, converging on one schema so a figure read from a photograph is indistinguishable downstream from one read from text.

Three decisions that shape the system

- **The model reads, code calculates.** Every margin, growth rate, EBITDA and runway figure is deterministic Python with its formula stored beside the result. An LLM silently miscalculating inside a board report is the failure mode this architecture exists to prevent.
- **“Not meaningful” is a first-class result.** Senus is loss-making, so ROCE, DSCR and EBITDA margin are computable and economically meaningless. Twelve values render as n/m with the reasoning retained, rather than printing a number that looks like analysis and is noise.
- **AI commentary is grounded in two sources only** — the already-computed metrics and management's own results-presentation transcript, never the raw filings. Every citation the model makes is checked server-side against what it was actually given before the UI renders it.

How outputs were validated

88 facts were hand-transcribed from source documents before any pipeline code existed, and a self-consistency checker proves that ground truth articulates (P&L foots, balance sheet balances). The extraction pipeline scores 100% precision and recall against it. Along the way, reconciliation caught a genuine €1,000 footing error in Senus's own published half-year results (turnover less cost of sales does not equal the printed gross profit) and a consolidated-vs-parent-company ambiguity in the statutory accounts — both are recorded exactly as printed and surfaced in the report, never silently corrected.

Stack & deployment

Python (pypdf, Pillow, Pydantic) for extraction and metrics · PostgreSQL for fact-level provenance · Next.js 16 / React 19 / TypeScript / Tailwind for the application · Claude Opus 5 with structured outputs throughout. Deployed to Vercel; the published report renders from a versioned, content-hashed payload with no runtime dependency on the database, confirmed live.

Links

Live app: senus-board-report-phi.vercel.app · Repository: github.com/seangavigan00-Obair/senus-board-report (private — access on request) · Full README covers architecture, validation methodology, assumptions and AI-assisted workflow in depth.